Warmup

1. Factor the polynomial, then sketch the graph. $f(x) = x^5 - 9x^3$.
Identify any local zeroes, local extrema, and the end behavior of the function.

2. Find the quotient and remainder using long division.
$f(x) = \frac{x^3+2x^2-x+1}{x+3}$

3. Use the remainder theorem to calculate P(7):
$P(x) = 6x^7 + 40x^6 + 16x^5 - 60x^3 - 69x^2 + 13x - 139$